

Mid

✓ Symmetric Cryptography

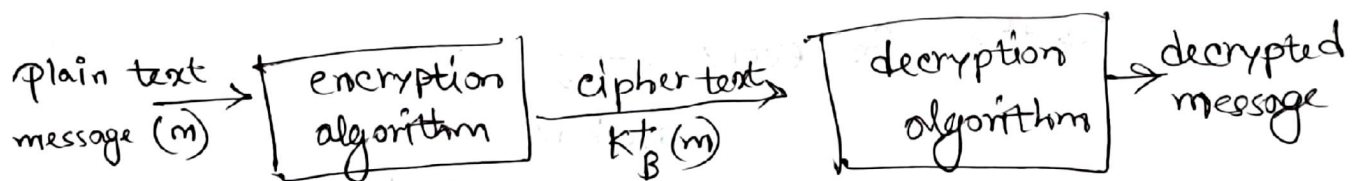
✓ Encryption-Decryption

Asymmetric Cryptography

Symmetric Key crypto: [sender, receiver] identical

asymmetric " " : [encryption public key]
public " " : [decryption secret key]

Public Key Cryptography



RSA Encryption Algorithm

RSA? → self study

RSA → public & private key

choosing keys:

- ✓ ① Choose two large prime numbers (1024 bits each)
- ✓ ② Compute $n = pq$, $z = (p-1)(q-1)$ $n \& z = 1024^2 \text{ bits}$
- ✓ ③ Choose e ($\because e < n$) that has no common factors with z (e, z are relative primes)
- ④ Choose d such as $ed-1$ is exactly divisible by z →
- ⑤ public key is (n, e) private key is (n, d) $ed \% z = 1$

K_T^+ K_B^-

4, 7 $\rightarrow$ relatively prime

$$\left. \begin{array}{l} 4 = 2 \times 2 \\ 7 = 7 \times 1 \end{array} \right\} \rightarrow \text{no common factors}$$

2, 7 $\rightarrow$ absolute prime

$$2 = 2 \times 1$$

$$7 = 7 \times 1$$

let $\boxed{ed-1}$

$\hookrightarrow$ such a way

we have $\textcircled{e}$ and $\textcircled{z}$

$\downarrow$
relatively
many

$\downarrow$
absolute

$\hookrightarrow$ the for every $e \rightarrow$

we have

$(ed - e)$ possibilities

RSA Encryption, Decryption

0. (n, e) & (n, d) as computed K_B^+ , K_B^-

0... $\triangleleft$ PDF ((section-26-8.1-8.2.ppt))

RSA example:

Bob chooses $p=5, q=7$ Then $n=35, \phi=24$

$e=5$ ($\because e, z$ relatively prime)

$d=29$ (so, $ed-1$ exactly divisible by z)

depan.

encrypt: $\frac{\text{letter}}{j} \quad \frac{m}{12} \quad \frac{m^e}{1524832} \quad \frac{c = m^e \bmod n}{17}$

decrypt: $\frac{c}{17} \quad \frac{c^d}{4.8196 \times 10^{35}} \quad \frac{m = c^d \bmod n}{12} \quad \frac{\text{letter}}{j}$

figure: process of encrypting
& decrypting 1 single letter.

Practice Problem